

LIFETIME ATHLETE

Personal Training + Wellbeing Tracker

Build specification for Cursor AI

Next.js + Supabase | Mobile-first PWA | Data-driven progression engine

Purpose

This document is the source of truth for building a personal workout, mobility, skill and performance tracker around the training plan already defined. It is written so Cursor can implement the product directly, while keeping exercise prescriptions, progression rules and training history separate from UI code.

Primary product goal

Open the app, immediately see exactly what is due today, start it with one tap, be guided set-by-set with the correct rest timers and technique notes, record completion with minimal friction, then automatically receive the next exact prescription based on exercise-specific progression rules.

User profile: 25-year-old experienced generalist athlete. Primary modality: calisthenics. Current focus skills: one-arm handstand, planche and splits. Broader goals: lifelong strength, endurance, mobility, agility, stability, explosiveness, power, swimming, breath-control competence and climbing, while minimizing avoidable pain and loss of mobility from inactivity.

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Important implementation rule

Do not hard-code the plan inside React components. Seed it into Supabase and render the experience from structured workout, exercise, set and progression data. The user must be able to edit routines later without a code change.

1. Product vision and non-negotiable principles

The app is a personal operating system for physical capability, not a generic gym log. It must combine scheduled training, skill practice, mobility, recovery, progression and history in one simple flow. The interface should make the correct action obvious without exposing database complexity.

Principle	Required behavior
Today first	The home screen opens on Today. The primary CTA is the next due routine or workout. No hunting through menus.
One step at a time	During a session, show the current exercise and current set prominently; next steps are visible but secondary.
Exact prescription	Every active exercise shows exact variant, sets, reps/seconds/distance/load and rest time. No vague ranges in the session player.
Skills while fresh	Workout ordering must preserve warm-up -> OAHS -> planche -> other skills -> power -> strength/endurance -> flexibility.
Automatic but explainable progression	After each exercise, completion data updates the next prescription using a named progression rule. Always show what changed and why.
Progress without reckless overload	High-skill/tendon-heavy movements use success credits or consecutive-success thresholds; a single good day may update state without forcing a harder variant immediately.
History is first-class	Every performed set, hold, load, duration, distance, completion status and progression event is stored.

Principle	Required behavior
Maintenance is not endless overload	Daily mobility can progress in range and quality while keeping the same low volume. The app must not add repetitions forever to maintenance work.
Mobile-first and interruption-proof	Timers and session state survive locking the phone, switching tabs, refreshes and temporary loss of network.
User remains in control	Every automatic change has Undo and Edit. The user can manually adjust a starting level, load, pool length, equipment increment or schedule.

1.1 Product scope

- Personal account and secure data storage in Supabase.
- Calendar-based training schedule with recurring weekly and 4-week-cycle logic.
- Workout/routine player with sets, reps, holds, distances, rest timers, notes and expandable exercise instructions.
- Per-exercise completion prompts and actual performance logging.
- Rule-based next-prescription generation.
- Progression history and charts by exercise, skill and capability domain.
- Mobility and split-depth tracking.
- Running, swimming and climbing support.
- Daily wellbeing/readiness logging and simple pain-aware safeguards.
- Installable mobile-first PWA behavior with offline session continuity.
- Plan editor so exercises, rules and schedules can be changed later.

1.2 Out of scope for first release

- Social feed, coaching marketplace or public leaderboards.
- Automatic medical diagnosis.
- Automatic camera-based form grading.
- Wearable integration. Keep the data model extensible for future Health/Garmin/Apple Health import, but do not block v1 on it.

2. UX and information architecture

2.1 Navigation

Primary tab	Purpose
Today	Next actions, daily maintenance, scheduled session, recovery state, week/cycle position.
Plan	Weekly calendar, 4-week cycle, routine library, plan editor.
Progress	History, skill metrics, strength/load trends, running/swimming/climbing metrics, adherence.
Library	Exercise explanations, progressions and technique notes.
Settings	Equipment increments, pool length, units, timer sounds/haptics, schedule, profile, export.

On mobile, use a fixed bottom navigation bar. On desktop, use a slim left sidebar. The visual hierarchy should remain identical across form factors.

2.2 Visual design direction

Area	Specification
Style	Clean, athletic, calm and technical. Avoid bodybuilding-app clutter and excessive motivational graphics.
Color	Neutral light/dark surfaces with one strong blue accent. Green = completed/on target; amber = needs attention; red reserved for pain/safety and failures.
Cards	Large touch targets, rounded cards, strong numeric hierarchy, generous spacing.
Typography	Large current-exercise title, very large set/rep or timer number, small secondary metadata.
Motion	Subtle transitions only. Timer completion may pulse; exercise completion may animate a checkmark.
Dark mode	Required. Gym and evening use are common.
One-hand use	Primary buttons should remain reachable near the bottom of the phone screen.

2.3 Today screen wireframe specification

TODAY Week 2 of 4
Good evening 72% ready

[DAILY MAINTENANCE] 5 min [Start]
Neck + shoulders + spine + hips + ankles

[PRIMARY SESSION] 90 min [Start workout]
Monday - Strength + Power
OAHS > Planche > Flag > Power > Strength > Splits

[NEXT SHORT SESSION]
Tuesday - OAHS + Planche + Front split / mobility

This week: 2 / 6 sessions completed [View week]
Current focus: OAHS | Planche | Splits

- If a session is overdue, label it Overdue but never shame the user.
- If two items are due, show the higher-priority training session first and maintenance second.
- If it is Sunday, show Recovery Day and the 5-minute maintenance routine only.
- Show cycle status such as Week 4 - Deload prominently because prescriptions change.

3. Core user journeys and screen specifications

3.1 First-run onboarding

The onboarding exists to convert unknown personal starting values into exact prescriptions. Do not ask the user for data that the training plan already defines.

Step	Fields / action
Account	Email/password or supported Supabase Auth method.
Units	kg, km, metres, seconds; default metric.
Equipment	Smallest load increment available; default 1.0 kg. Access to pull-up bar, dip bars, rings, weight belt/vest, pool, climbing gym.
OAHS level	Choose current assistance stage: wall weight shift / five-finger / three-finger / one-finger / free straddle / free legs-together.

Step	Fields / action
Planche level	Tuck / advanced tuck / one-leg / straddle / full. Default from current clean hold test.
Human flag level	One-leg / straddle / full. Initial state: one-leg.
HSPU level	Wall HSPU / freestanding / 2 cm deficit / 4 cm deficit / 6 cm deficit.
Strength loads	Starting external load for weighted pull-up 4x5 and Bulgarian split squat 3x8/leg.
Running	Comfortable steady pace and current 8 km time if known. If unknown, leave target pace blank and use duration/RPE for first cycle.
Splits	Front-left floor gap cm, front-right floor gap cm, middle-split floor gap cm; allow Unknown.
Swimming	Pool length: 25 m or 50 m. Default 25 m.
Climbing	Current comfortable bouldering grade and project grade; allow Unknown.
Schedule	Default Mon/Wed/Sat primary workouts; Tue/Thu/Fri short routines; Sun recovery. Allow changing days without changing sequence.

3.2 Starting a session

1. Tap Start from Today or Plan.
2. Create a training_session row and snapshot the prescription into session items. A session must never mutate underneath the user if the template changes later.
3. Show a 10-second overview: duration, focus, exercises, current deload status, and any pain/readiness warnings.
4. Tap Begin. Start at exercise 1. The app never forces the user to scroll through the entire workout during execution.
5. Allow Skip, Substitute only if explicitly configured, or End Session. Record all such events.

3.3 Exercise details drawer

The exercise name opens a bottom sheet or drawer. It must not navigate away from the active session.

- Plain-language setup and execution instructions.
- 3-5 key technique cues.
- Common mistakes.
- Purpose / capability trained.
- Current progression level and what unlocks the next level.
- Optional image/video field for future media.
- Pain caution specific to the exercise where relevant.

3.4 Completion prompt after every exercise

WEIGHTED PULL-UP COMPLETE
 Prescription: 4 sets x 5 reps @ +20.0 kg, 150 s rest

Did you complete every prescribed rep with clean form?
 [Yes, complete] [No / partial]

Optional: Difficulty Easy / Right / Hard
 Optional note: _____

If Yes is tapped, apply the exercise progression rule immediately and show a small Next time card, for example: Next Monday: 4 x 5 @ +21.0 kg. If the rule uses a threshold, show Progression credit 1 of 2 rather than pretending nothing happened. If No/partial is tapped, collect actual reps/seconds/load and apply the failure rule.

4. Session player and timer behavior

4.1 Session state machine

State	Behavior
planned	Scheduled but not started.
active	Current workout is in progress.
resting	A rest timer is active. This is still an active session sub-state.
paused	User intentionally paused; elapsed timer stops, but absolute timestamps are retained.
completed	User completed/end-reviewed session. Immutable snapshot except notes/corrections.
abandoned	Started but intentionally ended before completion. Preserve partial data.

4.2 Set interaction

1. Display Set 1 of N, exact target and current load/hold/distance.
2. User taps Complete set. For rep exercises, default actual reps to prescribed reps but allow quick +/- editing.
3. Immediately start the prescribed rest timer unless it is the final set.
4. Rest screen shows remaining time in large numerals, next set preview, Skip rest, +15 s and -15 s controls.
5. When timer reaches zero, play optional sound + vibration/haptic and show Start next set.
6. For holds or timed intervals, provide an optional built-in countdown/stopwatch, but still allow manual completion.
7. At the last set, show the exercise completion prompt, then progress to the next exercise.

Timer engineering requirement

Do not implement rest by decrementing an in-memory integer once per second. Store `rest_started_at` and `rest_duration_seconds`, calculate remaining time from `Date.now()`, persist active timer state locally, and restore it after tab suspension, screen lock or refresh.

4.3 Session summary

- Total time, completed/partial/skipped exercises.
- Every progression change generated during the session.
- Personal bests and skill best holds.
- Readiness vs perceived session difficulty.
- Pain/soreness notes.
- Preview of the next scheduled session.
- Undo any progression change before leaving the summary.

5. Training schedule: daily, weekly and monthly

5.1 Permanent daily maintenance - 5 minutes every day

Order	Exercise	Exact prescription
1	Chin tuck	8 reps
2	Neck rotation	5 reps per side
3	Wall slide	10 reps

Order	Exercise	Exact prescription
4	Cat-cow	8 reps
5	Open-book thoracic rotation	5 reps per side
6	90/90 hip switch	10 total
7	Deep squat hold	30 sec
8	Knee-over-toe ankle rock	10 reps per side

This routine is a maintenance routine. Completion improves adherence and movement quality but does not automatically add repetitions. Progression is range-of-motion quality: once all reps are pain-free and controlled, preserve the same dose.

5.2 Base week

Day	Duration	Session
Monday	90 min	Strength + Power. Skills first.
Tuesday	30 min	OAHS 5 min + light planche 5 min + front split/general mobility 20 min.
Wednesday	90 min	Athleticism + Endurance. Skills first.
Thursday	30 min	OAHS 5 min + light planche 5 min + deep flexibility 20 min.
Friday	30 min	OAHS 5 min + medium planche 5 min + joint/spine recovery 20 min, except swimming weeks.
Saturday	90 min	Calisthenics Volume + Work Capacity, except monthly climbing replacement.
Sunday	5 min + walking	Daily maintenance only. Full recovery day.

5.3 Fixed 4-week cycle

Week	Wednesday emphasis	Friday	Saturday	Loading
1	6 x (2 min strong + 2 min easy)	Normal 30-min recovery	Normal calisthenics volume	Normal
2	24 min continuous steady run	Performance swim, 900 m	Normal calisthenics volume	Progress
3	40 min continuous easy run	Normal 30-min recovery	Climbing session replaces normal Saturday	Highest workload
4	18 min very easy run	Easy technique swim, 800 m	Deload calisthenics	Deload: reduce sets 25-35%

In swimming weeks, Friday OAHS and planche are omitted, leaving five skill exposures that week. Sunday always remains recovery.

6. Exact routines and workouts

6.1 OAHS - 5-minute priority skill block (Monday-Saturday unless swim replaces Friday)

Step	Prescription	Rest / timer
Wrist rocks	10 forward + 10 lateral	Continuous
Finger/palm lifts	10 reps	Continuous
Chest-to-wall handstand weight shifts	2 x 20 sec	20 sec between holds
Fingertip-assisted OAHS	2 x 8 sec each arm, alternating	12 sec between attempts
Free OAHS attempts	2 attempts each arm, max 8 sec, alternating	15 sec between attempts

Quality rule: terminate any attempt when alignment deteriorates. Never turn the five-minute block into fatigue training. The app records best clean free hold for left and right separately.

6.2 Planche - 5-minute priority skill block

Day intensity	Exact work	Rest
Hard - Monday/Saturday	Planche lean 2 x 20 sec; current planche level 5 x 8 sec	30 sec between leans; 40 sec between holds
Medium - Wednesday/Friday	Planche lean 2 x 15 sec; current planche level 4 x 6 sec	30 sec between leans; 45 sec between holds
Light - Tuesday/Thursday	Planche lean 4 x 15 sec; current planche level 2 x 6 sec	30 sec between leans; 45 sec between holds

Planche level is a single global skill state used by all three prescriptions: tuck -> advanced tuck -> one-leg -> straddle -> full.

6.3 Monday - Strength + Power - 90 minutes

Block	Exercise	Exact prescription	Rest
Warm-up	Wrist rocks	10 forward + 10 lateral	No rest
Warm-up	Scapular push-ups	10	No rest
Warm-up	Scapular pull-ups	8	No rest
Warm-up	Arm circles	10 forward + 10 backward	No rest
Warm-up	Hollow-body hold	20 sec	No rest
Warm-up	Arch hold	20 sec	No rest
Warm-up	Air squat	15	No rest
Warm-up	Reverse lunge	8/leg	No rest
Skill	OAHS block	5 min protocol	As above
Skill	Planche - hard	5 min protocol	As above
Skill	One-leg human flag	3 x 6 sec each side	Left -> right -> 60 sec
Power	Strict muscle-up	4 x 3	90 sec
Power	Clap push-up	3 x 5	60 sec
Power	Standing broad jump	3 x 3	75 sec
Strength	Weighted pull-up	4 x 5 @ current load	150 sec

Block	Exercise	Exact prescription	Rest
Strength	Handstand push-up progression	4 x 5 @ current level	150 sec
Strength	Bulgarian split squat	3 x 8/leg @ current load	90 sec after both legs
Strength	Nordic hamstring curl	3 x 5, 3-sec eccentric	90 sec
Core	Hanging straight-leg raise	3 x 10	60 sec
Flexibility	Couch stretch	45 sec/side	10 sec transition
Flexibility	Half-split hamstring stretch	45 sec/side	10 sec transition
Flexibility	Front split	2 x 45 sec/side	20 sec between holds
Flexibility	Pancake	60 sec	20 sec
Flexibility	Middle split	60 sec	Finish

6.4 Wednesday - Athleticism + Endurance - 90 minutes

Block	Exercise	Exact prescription	Rest
Warm-up	Easy jog	2 min	-
Warm-up	Leg swings front/back	10/leg	-
Warm-up	Leg swings lateral	10/leg	-
Warm-up	Walking lunge	8/leg	-
Warm-up	Ankle pogo hop	20	-
Warm-up	High knees	20 sec	-
Warm-up	Butt kicks	20 sec	-
Skill	OAHS block	5 min protocol	As above
Skill	Planche - medium	5 min protocol	As above
Power	Pogo jump	3 x 12	45 sec
Speed	20 m acceleration sprint	4 x 20 m	90 sec
Agility	5-10-5 shuttle	4 reps, alternate start side	75 sec
Leg strength	Pistol squat	3 x 6/leg	90 sec
Posterior chain	Single-leg glute bridge	3 x 12/leg, 2-sec top hold	60 sec
Stability	Copenhagen plank	3 x 20 sec/side	45 sec after both sides
Lower leg	Single-leg calf raise	3 x 15/leg	Superset
Lower leg	Tibialis wall raise	3 x 20	45 sec after superset
Run - Week 1	Strong/easy intervals	6 x [2 min strong + 2 min easy] = 24 min	Continuous
Run - Week 2	Steady continuous run	24 min	Continuous
Run - Week 3	Easy continuous run	40 min; omit glute bridge + calf/tibialis blocks	Continuous
Run - Week 4	Very easy deload run	18 min	Continuous
Flexibility	Front split	45 sec/side	10 sec transition
Flexibility	Frog stretch	60 sec	10 sec
Flexibility	Pancake	60 sec	Finish

6.5 Saturday - Calisthenics Volume + Work Capacity - 90 minutes

Block	Exercise	Exact prescription	Rest
Warm-up	Wrist rocks	10 forward + 10 lateral	-
Warm-up	Scapular push-ups	10	-
Warm-up	Scapular pull-ups	8	-
Warm-up	Hollow hold	20 sec	-
Warm-up	Arch hold	20 sec	-
Warm-up	Bodyweight squat	15	-
Warm-up	Reverse lunge	8/leg	-
Warm-up	Shoulder circles	10 each direction	-
Skill	OAHS block	5 min protocol	As above
Skill	Planche - hard	5 min protocol	As above
Skill	One-leg human flag	3 x 8 sec/side	Left -> right -> 60 sec
Power	Strict muscle-up	4 x 3	90 sec
Volume superset A	Pull-up	4 x 10	Then dips
Volume superset A	Parallel-bar dip	4 x 12	90 sec after dip
Volume superset B	Inverted row	3 x 12	Then push-ups
Volume superset B	Push-up	3 x 20	60 sec after push-up
Legs	Walking lunge	3 x 12/leg	60 sec
Core	Dragon flag	3 x 6, 3-sec lowering	90 sec
Conditioning	Burpee	4 rounds x 8	Then mountain climbers
Conditioning	Mountain climber	4 rounds x 20 total	30 sec after round
Flexibility	Couch stretch	60 sec/side	10 sec transition
Flexibility	Half split	60 sec/side	10 sec transition
Flexibility	Front split	2 x 60 sec/side	20 sec between holds
Flexibility	Cossack squat	8/side	20 sec
Flexibility	Frog stretch	60 sec	20 sec
Flexibility	Pancake	60 sec	20 sec
Flexibility	Middle split	2 x 45 sec	20 sec

6.6 Tuesday - 30 minutes

Order	Exercise	Exact prescription	Rest
1	OAHS	5 min protocol	As above
2	Planche - light	5 min protocol	As above
3	Wall slide	2 x 10	20 sec
4	Open-book rotation	8/side	No rest
5	90/90 hip switch	2 x 10	20 sec
6	Couch stretch	2 x 45 sec/side	15 sec
7	Half split	2 x 45 sec/side	15 sec

Order	Exercise	Exact prescription	Rest
8	Front split	2 x 45 sec/side	20 sec
9	Dead bug	2 x 8/side	30 sec

6.7 Thursday - 30 minutes

Order	Exercise	Exact prescription	Rest
1	OAHS	5 min protocol	As above
2	Planche - light	5 min protocol	As above
3	Cossack squat	2 x 8/side	20 sec
4	90/90 hip switch	10 total	No rest
5	Couch stretch	60 sec/side	15 sec
6	Half split	60 sec/side	15 sec
7	Front split	2 x 45 sec/side	20 sec
8	Frog stretch	2 x 60 sec	20 sec
9	Pancake	2 x 60 sec	20 sec
10	Middle split	2 x 45 sec	20 sec
11	Straight-leg seated lift	2 x 8/leg	30 sec

6.8 Friday - 30 minutes (normal weeks)

Order	Exercise	Exact prescription	Rest
1	OAHS	5 min protocol	As above
2	Planche - medium	5 min protocol	As above
3	Chin tuck	10	No rest
4	Wall slide	2 x 10	20 sec
5	Open-book rotation	8/side	No rest
6	Bird dog	2 x 6/side, 3-sec hold	30 sec
7	90/90 hip switch	2 x 8	20 sec
8	Knee-to-wall ankle mobilization	10/side	No rest
9	Calf stretch	45 sec/side	10 sec
10	Front split	30 sec/side	10 sec
11	Frog stretch	45 sec	10 sec
12	Pancake	45 sec	Finish

6.9 Week 2 Friday - performance swim - 900 m

Order	Work	Exact prescription	Rest
1	Easy freestyle	200 m	30 sec
2	Technique	4 x 50 m: 25 m catch-up + 25 m normal freestyle	20 sec

Order	Work	Exact prescription	Rest
3	Strong controlled freestyle	6 x 50 m	30 sec
4	Fast freestyle	4 x 25 m	40 sec
5	Easy freestyle	100 m	Finish

6.10 Week 4 Friday - recovery/technique swim - 800 m

Order	Work	Exact prescription	Rest
1	Easy freestyle	200 m	30 sec
2	Technique	4 x 50 m: 25 m catch-up + 25 m normal freestyle	20 sec
3	Freestyle breathing control	4 x 50 m breathing every 3 strokes	30 sec
4	Backstroke	4 x 25 m	20 sec
5	Very easy freestyle	100 m	Finish

Breath-hold safety

Do not prescribe unsupervised underwater breath-hold repetitions. Normal swim sessions may train breathing rhythm only. Dedicated apnea/freediving work requires a competent dedicated observer/instructor, no hyperventilation, and a separate safety protocol.

6.11 Week 3 Saturday - climbing replacement - 90 minutes

Block	Exercise	Exact prescription	Rest
Warm-up	Wrist circles	10 each direction	-
Warm-up	Finger open/close	20	-
Warm-up	Scapular pull-up	2 x 8	20 sec
Warm-up	Scapular push-up	2 x 10	20 sec
Warm-up	Arm circles	10 each direction	-
Skill	OAHS	5 min protocol	As above
Skill	Planche - hard	5 min protocol	As above
Climb warm-up	Easy problems/routes	2	60 sec
Climb warm-up	Slightly harder problems/routes	2	90 sec
Quality attempts	Problem A	3 attempts	2 min
Quality attempts	Problem B	3 attempts	2 min
Quality attempts	Problem C	3 attempts	2 min
Quality attempts	Problem D	3 attempts	2 min
Endurance	Continuous easy climbing	3 x 4 min	2 min
Antagonist	Push-up	2 x 20	45 sec
Antagonist	Band finger extension	2 x 20	30 sec
Flexibility	Couch stretch	45 sec/side	10 sec

Block	Exercise	Exact prescription	Rest
Flexibility	Half split	45 sec/side	10 sec
Flexibility	Front split	60 sec/side	10 sec
Flexibility	Frog stretch	60 sec	10 sec
Flexibility	Pancake	60 sec	10 sec
Flexibility	Middle split	60 sec	10 sec
Forearm	Flexor stretch	30 sec/side	-
Forearm	Extensor stretch	30 sec/side	Finish

6.12 Week 4 deload rules

- Any 4-set exercise becomes exactly 3 sets.
- Any 3-set exercise becomes exactly 2 sets.
- Repetitions per remaining set stay unchanged.
- OAHS remains full 5 minutes on scheduled OAHS days.
- Planche uses the light Tuesday/Thursday protocol on every scheduled planche day.
- Human flag becomes 2 sets.
- Saturday conditioning becomes 2 rounds.
- Wednesday run is exactly 18 minutes easy.
- Split work remains but intensity target is 5/10; do not chase maximum depth.

7. Exercise-specific progression engine

Progression must be deterministic and stored as data. The session player submits actual performance. A server-side progression function evaluates the rule attached to that routine item or global skill, records a `progression_event`, and updates the next prescription. The UI then shows the result.

7.1 Generic rule contract

```
type ProgressionRule = {
  scope: 'global_skill' | 'routine_item' | 'capability';
  successCriteria: object;
  onSuccess: ProgressionAction[];
  onFailure: ProgressionAction[];
  threshold?: { consecutiveSuccesses?: number; creditsRequired?: number };
  caps?: object;
  deloadBehavior?: object;
};

type ProgressionAction =
  | { type: 'increment'; metric: string; amount: number }
  | { type: 'decrement'; metric: string; amount: number }
  | { type: 'advance_level'; levelTable: string; reset: object }
  | { type: 'hold' }
  | { type: 'set_target_from_metric'; metric: string; formula: string };
```

Crucial UX rule

A Yes response must always update progression state. For some exercises the visible prescription changes immediately. For high-skill or tendon-heavy movements, Yes may add a progression credit; when the configured threshold is reached the app advances the visible level. This preserves the requested continuous adaptation without reckless day-to-day jumps.

7.2 Exact progression rules - priority skills and flexibility

Exercise/skill	Rule
OAHS	Record clean best free hold L/R and assistance level. Yes = +1 skill credit. After 3 successful practice sessions with both sides completing prescribed assisted attempts and at least one clean free attempt, advance one assistance level. Keep 5-minute duration unchanged. If already free straddle, progression = longer clean hold; then legs-together level. Failure resets consecutive-success counter only, never auto-regresses level.
Planche	Hard sessions drive level changes. Yes on a hard session = +1 hard-success credit. After 2 successful hard sessions at all 5 x 8 sec holds with clean protraction/locked elbows, advance tuck -> advanced tuck -> one-leg -> straddle -> full. New level resets hard holds to 5 x 5 sec, medium to 4 x 4 sec, light to 2 x 4 sec; each successful hard session adds 1 sec/hold until original 8/6/6 targets are restored. Three failed hard sessions reduce hold duration by 20%, not level.
Human flag	Monday and Saturday prescriptions keep independent target seconds but share one level: one-leg -> straddle -> full. Every successful session adds +1 sec to that workout context, cap 10 sec. When both Monday and Saturday targets reach 10 sec and are completed once, advance level and reset Monday to 5 sec, Saturday to 6 sec.
Front split	Track left/right support height or floor gap in cm. A successful moderate/deep split session earns 1 credit. After 2 credits, reduce target support/floor gap by exactly 1 cm for that side, hold times unchanged. Never progress below 0 cm.
Middle split	Track floor gap/support height. After 2 successful deep sessions, reduce target by 1 cm. Hold times unchanged.
General mobility	Maintenance progression only. Keep exact reps. Once full pain-free ROM is achieved, mark quality = maintained; do not add volume.

7.3 Exact progression rules - Monday

Exercise	Rule
Strict muscle-up 4x3	After 2 successful sessions with all 12 reps clean and no grind, add +1.0 kg external load next time. Keep 4x3 and 90 sec rest. Failure: keep load. Two failures at same load: remove 1.0 kg.
Clap push-up 3x5	Success: keep 3x5 and increase quality target, not fatigue volume. Track optional estimated airtime/height. After 3 successful sessions, progress exercise level to higher explosive push-up variation selected in library; reset success counter. Rest remains 60 sec.
Broad jump 3x3	Record best distance and median of 9 jumps. Success = all 9 quality landings. Next target = prior median x 1.01, rounded to nearest cm. Sets/reps/rest unchanged.
Weighted pull-up 4x5	Every successful session: +1.0 kg next Monday. Failure: repeat same load. Two consecutive failures: -2.0 kg and reset failure counter.
HSPU progression 4x5	Success: next session becomes 4x6. Next success becomes 4x7. Next success advances one level: wall -> freestanding -> 2 cm deficit -> 4 cm deficit -> 6 cm deficit, and resets to 4x5. Failure: repeat exact prescription.
Bulgarian split squat 3x8/leg	Success: +2.0 kg total external load next Monday. Failure: hold load. Two consecutive failures: -2.0 kg.
Nordic curl 3x5	Success reduces assistance one defined level. If unassisted, eccentric target goes 3 sec -> 4 sec -> 5 sec. At 5-sec unassisted, progress to 3x6 and then 3x7; after 3x7, add 1.0 kg and reset 3x5.
Hanging straight-leg raise 3x10	Success -> 3x11, then 3x12. After successful 3x12, advance to toes-to-bar 3x8. Thereafter 3x9 -> 3x10 -> add 0.5 kg ankle load and reset 3x8.

7.4 Exact progression rules - Wednesday

Exercise	Rule
Pogo jump 3x12	Keep volume fixed. Success for 3 sessions advances stiffness/height target one library level. Rest stays 45 sec.
20 m sprint 4 reps	Record each time. Success = all four completed and slowest $\leq 103\%$ of fastest. Next target time = current best $\times 0.995$. Volume/rest unchanged.
5-10-5 shuttle 4 reps	Record each time. Success = all four completed and slowest $\leq 104\%$ of fastest. Next target = current best $\times 0.995$.
Pistol squat 3x6/leg	Success: +1.0 kg external load next Wednesday. Failure: repeat.
Single-leg glute bridge 3x12/leg	Success $\rightarrow 3x13 \rightarrow 3x14 \rightarrow 3x15$. After 3x15, progress to feet-elevated 3x12 and repeat sequence.
Copenhagen plank 3x20 sec/side	Success: +5 sec/side next time up to 30 sec. After 30 sec, advance to longer-lever variation and reset to 20 sec.
Single-leg calf raise 3x15	Success adds +1 rep/set until 3x20. Then add +5.0 kg and reset to 3x15.
Tibialis wall raise 3x20	Success adds +2 reps/set to 3x30. Then move feet 5 cm farther from wall and reset 3x20.
Week 1 intervals	If all 6 fast intervals completed and session difficulty $\leq 8/10$, next cycle fast-interval pace target improves by 5 sec/km. Durations stay 2 min fast/2 min easy.
Week 2 steady run	Success at difficulty $\leq 7/10$: next cycle duration +2 min, until 30 min. At 30 min, keep duration and improve target pace by 5 sec/km per successful cycle.
Week 3 long easy run	Success: next cycle +5 min until 60 min. At 60 min, keep duration and improve target pace by 5 sec/km per successful cycle.
Week 4 easy run	No progression. It is a deload/recovery prescription.

7.5 Exact progression rules - Saturday

Exercise	Rule
Pull-up 4x10	Success $\rightarrow 4x11 \rightarrow 4x12$. After 4x12, add +2.0 kg and reset to 4x8; then 4x9 $\rightarrow 4x10 \rightarrow 4x11 \rightarrow 4x12$ before next +2.0 kg.
Dip 4x12	Success $\rightarrow 4x13 \rightarrow 4x14 \rightarrow 4x15$. Then add +2.0 kg and reset to 4x10.
Inverted row 3x12	Success $\rightarrow 3x13 \rightarrow 3x14 \rightarrow 3x15$. Then advance body-angle/feet-elevation level and reset 3x12.
Push-up 3x20	Success adds +2 reps/set: 22, 24, 26, 28, 30. After 3x30, add +5.0 kg vest/backpack and reset to 3x20.
Walking lunge 3x12/leg	Success $\rightarrow 13 \rightarrow 14 \rightarrow 15$ reps/leg. After 15, add +4.0 kg total and reset to 12.
Dragon flag 3x6	Success $\rightarrow 3x7 \rightarrow 3x8$. Then add +0.5 kg ankle load and reset 3x6.
Burpee + mountain climber circuit	Success: reduce round rest 30 $\rightarrow 25 \rightarrow 20 \rightarrow 15$ sec. At 15 sec and another success, reset rest to 30 sec and change each round to 9 burpees + 22 mountain climbers. Continue same pattern to cap 12 burpees + 28 climbers, then hold.

7.6 Swimming and climbing progression

Session	Rule
Week 2 performance swim	If all work completed and difficulty $\leq 8/10$, add exactly one 50 m strong freestyle repeat next performance swim, increasing total by 50 m. Continue until total session = 1200 m. Then reduce strong-repeat rest 30 -> 25 -> 20 -> 15 sec across successful sessions.
Week 4 recovery swim	No overload progression. Keep 800 m and improve technique quality only.
Climbing quality problems	Store grade for A-D. If a problem is completed within 1-2 of the 3 attempts, next month that problem increases by one grade step. If completed on attempt 3, repeat same grade. If not completed, repeat same grade.
Climbing endurance 3x4 min	If all three intervals completed without falling from pump, next month each interval +30 sec. Cap at 6 min. After successful 3x6 min, increase circuit grade one step and reset to 3x4 min.

7.7 Failure and pain-aware behavior

- No/partial completion must never be treated as a failure of the whole workout. Store actuals and apply only that exercise rule.
- If the user reports sharp pain, radiating pain, numbness/tingling or weakness, do not auto-progress the affected exercise. Display a clear stop/assessment message rather than coaching through it.
- If local pain is rated $\geq 4/10$, freeze progression on relevant movements for that session and mark the result as pain-limited.
- If ordinary muscular difficulty is high but pain is 0-3/10, use normal success/failure rules.
- All automatic decisions are reversible from the session summary and exercise history.

8. Exercise explanation library

Each exercise record should contain short instructions suitable for an expandable drawer. Seed the following text. Cursor may create the records through a migration/seed script.

Exercise	How to perform	Key cues	Common mistake
Chin tuck	Stand or sit tall. Glide the head straight backward without looking down.	Keep eyes level; make a gentle double-chin; no force.	Jutting the chin downward or extending the neck.
Neck rotation	Turn the head slowly left and right through comfortable range.	Stay tall; shoulders relaxed; move slowly.	Forcing end range or rotating the torso.
Wall slide	Back/forearms against wall where possible. Slide arms upward while keeping ribs controlled.	Do not shrug; keep rib cage down; move pain-free.	Arching the lower back to fake overhead range.
Cat-cow	On hands and knees, alternate controlled spinal flexion and extension.	Move segment by segment; breathe normally.	Throwing the head or forcing lumbar extension.
Open-book rotation	Side-lying with hips/knees bent, rotate top arm and chest open.	Keep knees together; rotate through thoracic spine.	Letting the pelvis roll excessively.
90/90 hip switch	Sit with both knees bent at 90 degrees and rotate them side to side.	Stay tall; control the transition.	Using momentum or forcing painful hip range.
Deep squat hold	Sit into a comfortable deep squat with feet planted.	Knees track toes; heels stay down; torso relaxed.	Collapsing onto painful joints.
Knee-over-toe ankle rock	Drive knee forward over toes while heel stays planted.	Track knee in line with toes.	Heel lifting or foot collapsing.

Exercise	How to perform	Key cues	Common mistake
Scapular push-up	In plank, keep elbows straight and move shoulder blades apart/together.	Push the floor away; ribs controlled.	Bending elbows.
Scapular pull-up	Hang with straight elbows; depress/retract shoulder blades to lift body slightly.	Long arms; small controlled movement.	Turning it into a partial pull-up.
Hollow hold	Lie supine, flatten lower back, lift shoulders and legs.	Lower back stays pressed down.	Arching lumbar spine.
Arch hold	Lie prone and lift chest/legs slightly with long body line.	Glutes on; neck neutral.	Cranking neck or lumbar spine.
OAHS wall weight shift	In a chest-to-wall handstand, shift shoulders and hips over one hand while reducing pressure through the other.	Keep pushing tall through support shoulder; stacked line.	Twisting pelvis or collapsing shoulder.
Fingertip-assisted OAHS	Balance mainly on one hand while opposite fingertips provide minimal assistance.	Reduce fingertip pressure gradually; keep hips over base.	Using the assisting hand as a second full support.
Free OAHS attempt	Enter a stable handstand and transfer balance onto one hand.	Stay tall; control hip position; exit safely when line is lost.	Long fatigue saves or repeated uncontrolled falls.
Planche lean	From straight-arm support, protract shoulders and lean shoulders forward past wrists.	Locked elbows; rounded/protracted upper back; body tight.	Bent elbows or sagging hips.
Planche hold	Hold the currently assigned planche progression with straight elbows and protracted scapulae.	Push floor away; posterior pelvic tilt; stop when form breaks.	Counting seconds after elbows bend or hips drop.
One-leg human flag	Grip vertical support, press with lower arm and pull with upper arm while extending one leg.	Shoulders active; pelvis stacked; train both sides.	Twisting chest or hanging passively on shoulder.
Strict muscle-up	From controlled hang, pull explosively high, transition chest over bar, finish with straight-arm dip support.	Keep path close to bar; no uncontrolled kip.	Chicken-wing transition or grinding slow reps.
Clap push-up	Lower under control and push explosively enough to clap before landing softly.	Rigid trunk; hands return under shoulders.	Sagging hips or stiff-elbow landing.
Standing broad jump	Countermove, swing arms and jump forward, landing quietly in stable squat.	Full reset each rep; stick landing.	Repeated rebound jumps or unstable landing.
Weighted pull-up	Start from controlled hang and pull until chin clearly passes bar.	Depress shoulders; no leg swing; full controlled lower.	Half reps or kipping.
Handstand push-up	From handstand, lower head under control and press back to lockout using current progression level.	Vertical line; active shoulders; controlled depth.	Flaring excessively or losing trunk line.
Bulgarian split squat	Rear foot elevated, lower front leg until controlled depth, then stand.	Front knee tracks toes; torso stable; drive through whole foot.	Pushing excessively off rear leg.
Nordic hamstring curl	Kneel with ankles anchored, keep hips extended and lower body forward under hamstring control.	Straight line knee-to-shoulder; use assistance as prescribed.	Bending at hips to shorten lever.
Hanging straight-leg raise	Hang and raise straight legs using abs/hip flexors without swinging.	Posteriorly tilt pelvis at top; controlled lower.	Kipping or momentum.

Exercise	How to perform	Key cues	Common mistake
Pogo jump	Bounce vertically from ankles with small knee bend and short ground contact.	Tall posture; springy ankles; quiet landing.	Deep squat or slow contacts.
20 m acceleration sprint	Accelerate hard from a stable start through 20 m.	Forward projection early; powerful arms; full recovery.	Turning reps into fatigued conditioning.
5-10-5 shuttle	Sprint 5 m to one side, 10 m across, then 5 m through start.	Lower center of mass before cuts; plant outside foot cleanly.	Standing upright into change of direction.
Pistol squat	Single-leg squat to controlled depth and return without assistance unless current level requires it.	Knee tracks toes; heel planted; torso controlled.	Knee collapsing inward or bouncing off bottom.
Single-leg glute bridge	Lie supine, one foot planted, extend hip and pause at top.	Ribs down; squeeze glute; pelvis level.	Hyperextending lumbar spine.
Copenhagen plank	Side plank with top leg supported on bench/chair and lower leg free.	Straight body; hips high; shoulder stacked.	Letting hips rotate or drop.
Single-leg calf raise	Rise fully onto forefoot on one leg and lower under control.	Full range; stable ankle.	Bouncing through reps.
Tibialis wall raise	Lean on wall with heels planted and lift toes toward shins.	Keep heels down; pause at top.	Rocking whole body.
Pull-up	Controlled hang to chin over bar and back to full extension.	No swing; consistent ROM.	Kipping or shortened lower.
Parallel-bar dip	From straight-arm support, lower shoulders below elbows as comfortable and press to lockout.	Shoulders active; controlled depth.	Dropping fast or shrugging at top.
Inverted row	Body straight under bar/rings; pull chest to hands/bar.	Squeeze shoulder blades; rigid body.	Hips sagging.
Push-up	Body straight, lower chest near floor, press to full lockout.	Elbows controlled; ribs and pelvis aligned.	Sagging hips or partial depth.
Walking lunge	Step forward into lunge, lower under control and drive into next step.	Front knee tracks toes; torso stable.	Short unstable steps or knee collapse.
Dragon flag	Brace on bench/support, raise rigid body and lower slowly as one unit.	Posterior pelvic tilt; controlled 3-sec lower.	Bending hips to make rep easier.
Burpee	Squat to hands, move feet to plank, return and stand/jump.	Smooth rhythm; stable plank.	Losing trunk control under fatigue.
Mountain climber	From plank, alternate driving knees forward.	Shoulders over hands; hips stable.	Bouncing hips high.
Couch stretch	Rear knee near wall/bench, rear foot elevated, squeeze glute and bring torso upright.	Posterior pelvic tilt; glute on.	Arching lumbar spine.
Half split	From kneeling lunge, shift hips back and straighten front leg.	Long spine; hinge from hips.	Rounding aggressively to reach farther.
Front split	Slide front heel forward and rear leg back while keeping hips as square as possible.	Use supports; breathe; no sharp pain.	Forcing depth or twisting hips open.
Cossack squat	Shift into deep side squat while opposite leg stays straight.	Working foot planted; knee tracks toes.	Collapsing knee inward.
Frog stretch	On forearms/hands and knees, widen knees and move hips back gently.	Neutral spine; controlled range.	Forcing groin stretch sharply.

Exercise	How to perform	Key cues	Common mistake
Pancake	Sit legs wide and hinge torso forward from hips.	Long spine; active hip flexion.	Rounding only through lower back.
Middle split	Slide legs apart laterally with support, controlling depth.	Knees/toes positioned consistently; use blocks.	Forcing depth or bouncing.
Straight-leg seated lift	Sit tall with one/both legs straight and actively lift heel off floor.	Keep knee locked; use hip flexors.	Leaning back excessively.
Dead bug	Supine with hips/knees 90 degrees; extend opposite arm/leg without losing trunk position.	Lower back controlled; slow reps.	Arching as limb extends.
Bird dog	On hands/knees, extend opposite arm and leg and hold.	Pelvis square; reach long.	Rotating trunk.
Catch-up freestyle drill	Swim freestyle with one hand waiting forward until the other hand reaches it before the next stroke.	Long body line; relaxed kick; focus on catch timing.	Pausing so much that body sinks.
Freestyle breathing every 3 strokes	Swim normal freestyle and breathe every third arm stroke, alternating sides.	Exhale continuously underwater; relaxed breath.	Holding breath underwater or forcing pace.
Climbing quality attempt	Make one deliberate attempt on the assigned problem with full focus, then rest.	Read sequence first; precise feet; use open-hand/half-crimp where practical.	Rapid repeated attempts without recovery.
Continuous easy climbing	Stay on easy terrain continuously for assigned time.	Efficient feet; relaxed grip; straight arms when possible.	Choosing terrain too hard to sustain.
Band finger extension	Open fingers against an elastic band around fingertips.	Full opening; controlled return.	Snapping band or rushing.

9. Supabase data model and security

Use PostgreSQL as the source of truth. The live session may be cached locally for interruption resistance, but completion must sync to Supabase. Use UUID primary keys, created_at/updated_at timestamps, and user_id on all user-owned tables.

9.1 Core tables

Table	Purpose / important fields
profiles	user_id PK/FK auth.users, display_name, units, timezone, smallest_load_increment_kg, pool_length_m
capability_goals	id, user_id, code, name, priority, target_text, active
exercise_library	id, slug, name, category, metric_type, instructions, cues, common_mistakes, media_url, progression_family
exercise_levels	id, exercise_id, ordinal, name, parameters jsonb, next_level_id
routine_templates	id, user_id nullable for system seed, name, kind, default_duration_min, description, active
routine_items	id, routine_template_id, exercise_id, sequence, block, prescription jsonb, rest_seconds, progression_rule_id, progression_scope
training_cycles	id, user_id, cycle_number, start_date, end_date, status
scheduled_sessions	id, user_id, date, routine_template_id, cycle_week, day_role, status, generated_from_schedule
training_sessions	id, user_id, scheduled_session_id, routine_template_id, started_at, ended_at, status, readiness_snapshot jsonb, notes

Table	Purpose / important fields
session_items	id, training_session_id, routine_item_id, exercise_id, sequence, prescription_snapshot jsonb, progression_state_snapshot jsonb, status
set_results	id, session_item_id, set_index, prescribed jsonb, actual jsonb, started_at, completed_at, success, note
exercise_results	id, session_item_id, completed_all boolean, difficulty, pain_score, note, metrics jsonb
progression_rules	id, code, name, version, rule_json jsonb, description
progression_states	id, user_id, scope_type, scope_id, current_level_id, state jsonb, success_credits, consecutive_successes, consecutive_failures
progression_events	id, user_id, session_item_id, rule_id, event_type, before jsonb, after jsonb, explanation, created_at, undone_at
wellbeing_checkins	id, user_id, date, sleep_quality_1_5, energy_1_5, soreness_0_10, neck_pain_0_10, back_pain_0_10, notes
performance_measurements	id, user_id, metric_code, value_numeric, unit, side, measured_at, context jsonb
user_schedule_preferences	id, user_id, weekday_map jsonb, reminder_preferences jsonb

9.2 Prescription JSON shape

```
{
  "sets": 4,
  "reps_per_set": 5,
  "hold_seconds": null,
  "distance_m": null,
  "duration_seconds": null,
  "load_kg": 20,
  "exercise_level_id": "uuid",
  "tempo": null,
  "rest_seconds": 150,
  "per_side": false,
  "notes": "Strict reps; no kip"
}
```

Use the same schema for snapshot data, but allow exercise-specific fields in an extras object. This lets the session player remain generic while still supporting sprints, swimming, climbing grades, split depth and timed skill holds.

9.3 Progression state examples

```
// Weighted pull-up scoped to Monday routine item
{
  "load_kg": 20,
  "consecutive_failures": 0
}

// Planche global skill state
{
  "level": "advanced_tuck",
  "hard_hold_seconds": 7,
  "medium_hold_seconds": 5,
  "light_hold_seconds": 5,
  "hard_success_credits": 1
}

// Front split left side
{
  "target_floor_gap_cm": 8,
  "success_credits": 1
}
```

9.4 RLS and security

- Enable Row Level Security on every exposed table.
- All user-owned rows must be readable/writable only where `auth.uid() = user_id`.
- System seed `exercise_library` and `progression_rules` can be readable to authenticated users but writable only through trusted server/admin pathways.
- Never expose Supabase service-role credentials to the browser.
- Use server-side actions/functions for progression evaluation so clients cannot silently forge progression state.
- Validate every mutation with Zod or an equivalent runtime schema.
- Prefer cookie-based Supabase Auth in the Next.js App Router using the current official `@supabase/ssr` pattern; do not hard-pin code to an obsolete API if the library changes.

```
-- Example pattern; repeat appropriately per table.
alter table public.training_sessions enable row level security;

create policy "users read own sessions"
on public.training_sessions for select
using (auth.uid() = user_id);

create policy "users create own sessions"
on public.training_sessions for insert
with check (auth.uid() = user_id);

create policy "users update own sessions"
on public.training_sessions for update
using (auth.uid() = user_id)
with check (auth.uid() = user_id);
```

10. Next.js architecture and application logic

10.1 Recommended stack

Layer	Recommendation
Framework	Next.js App Router + TypeScript. Use current stable release.
Styling	Tailwind CSS. Use a reusable component system such as shadcn/ui if desired; keep custom workout-player primitives lightweight.
Icons	Lucide or equivalent simple line icon set.
Validation	Zod for action payloads, database-derived forms and progression inputs.
Backend	Supabase Postgres + Auth. Storage only if exercise media is uploaded later.
Charts	Lightweight charting library such as Recharts; lazy-load Progress views.
Local session state	A small persisted client store (for example Zustand) or equivalent. Persist only active session/timer draft; Supabase remains source of truth.
Testing	Vitest/Jest for progression rules; Playwright for critical workout flows.

10.2 Route structure

```
/app
  /(auth)/login
  /(app)/today
  /(app)/plan
  /(app)/plan/[routineId]
```

```

/(app)/session/[sessionId]
/(app)/session/[sessionId]/summary
/(app)/progress
/(app)/progress/[exerciseSlug]
/(app)/library
/(app)/library/[exerciseSlug]
/(app)/settings
/api (only where Route Handlers are genuinely useful)
/lib
/supabase/client.ts
/supabase/server.ts
/progression/
/training/
/validation/
/components
/workout-player/
/timers/
/progress/
/ui/

```

10.3 Server vs client responsibilities

Server	Client
Auth/session validation	Live timer rendering
Today/plan data fetch	Active workout interaction
Create session and snapshot prescription	Set completion controls
Commit set/exercise results	Local offline queue
Evaluate progression rules transactionally	Audio/haptic feedback
Generate future scheduled sessions	Optimistic UI and undo presentation
History/aggregate queries	Drawer/modals and charts interaction

10.4 Progression transaction

1. Client submits exercise completion payload with exact actuals and idempotency key.
2. Server validates ownership, session status and payload schema.
3. Insert/update set_results and exercise_result.
4. Lock/read current progression_state for the relevant scope.
5. Evaluate progression rule deterministically.
6. Insert progression_event containing full before/after JSON and human explanation.
7. Update progression_state.
8. Commit transaction.
9. Return nextPrescriptionPreview to client.
10. If user taps Undo from summary, set progression_event.undone_at and restore before state in a new audited transaction.

10.5 Schedule generation

Generate scheduled_sessions at least 6 weeks ahead. The schedule engine assigns the correct week-of-cycle variant. Moving a session to another date should preserve its cycle role unless the user explicitly chooses Reschedule and reflow week.

- Monday: Strength + Power.
- Tuesday: OAHS + light planche + front split/mobility.
- Wednesday: Athleticism + Endurance with week-specific run.
- Thursday: OAHS + light planche + deep flexibility.

- Friday: normal recovery except Week 2 performance swim and Week 4 recovery swim.
- Saturday: normal volume except Week 3 climbing; Week 4 deload Saturday.
- Sunday: recovery + maintenance only.

11. History, analytics and wellbeing

11.1 Progress dashboard

Card/chart	Data
Consistency	Sessions completed / scheduled over 7, 28 and 90 days.
OAHS	Best clean hold L/R, assistance level timeline, successful practice streak.
Planche	Current level, hard-hold seconds, level-change history.
Human flag	Level + Monday/Saturday target seconds by side.
Strength	Weighted pull-up load, HSPU level/reps, Bulgarian load, Nordic level.
Power	Broad-jump best/median; 20 m sprint best; shuttle best.
Endurance	Run durations and pace; interval target pace; long-run trend.
Flexibility	Front-left/right and middle floor-gap cm, plotted downward toward zero.
Swimming	Total metres, strong-repeat count/rest, session completion.
Climbing	Problem grades and endurance interval duration.
Pain/wellbeing	Sleep/energy/soreness plus neck/back pain over time, with training overlay.

11.2 Exercise history page

- Current exact prescription at top.
- Next unlock / progression rule in plain language.
- Chart of the primary metric appropriate to the exercise.
- Chronological session list with prescribed vs actual values.
- Progression-event timeline: +1 kg, advanced to advanced tuck, split target -1 cm, etc.
- Ability to correct a logged result without deleting history; maintain audit timestamps.

11.3 Daily readiness

Keep this optional and fast: one tap per metric. It is for trend awareness, not diagnosis.

Metric	Scale
Sleep quality	1-5
Energy	1-5
General soreness	0-10
Neck pain	0-10
Back pain	0-10
Notes	Optional free text

The initial app should not automatically cancel training from a low readiness score. It may show a caution banner. Pain-aware progression rules described above may freeze progression for affected exercises.

12. Offline/PWA behavior, accessibility and polish

12.1 PWA behavior

- Installable manifest with app icon and standalone display.
- Cache the shell, exercise library text and today/current-session snapshot.
- If network drops mid-session, all set completions remain usable and enter a local mutation queue.
- When connectivity returns, sync in order using idempotency keys.
- Never lose an active timer because of network loss.
- Display a subtle Offline - changes will sync indicator.

12.2 Accessibility and gym usability

- Minimum 44x44 px touch targets.
- All critical state conveyed by text/icons as well as color.
- Large timer numerals and strong contrast.
- Keyboard support on desktop.
- ARIA labels for timer controls and exercise completion actions.
- Reduced motion preference respected.
- Sound and vibration can each be independently disabled.
- Prevent accidental session loss: back navigation prompts if there are unsynced changes.

12.3 Nice UX details that materially improve use

- Keep screen awake during active sessions if browser capability is available and user grants permission.
- Auto-scroll/focus the active set after a rest timer ends.
- Quick +/- buttons for actual reps and load instead of forcing keyboard entry.
- Long-press or menu for Mark exercise as pain-limited, Skip, or Edit prescription.
- Persistent mini rest timer if the user opens exercise instructions during rest.
- Celebrations are subtle: use PR badge or checkmark, not full-screen confetti every set.
- At the end of a 4-week cycle, generate a compact monthly review and the next cycle prescriptions.

13. Acceptance criteria and test plan

13.1 Must-pass product acceptance criteria

Area	Pass condition
Today	On any date, user can see exactly what is due and launch it in <=2 taps.
Session fidelity	Starting a routine snapshots the exact prescription so later plan edits do not alter an active/historical session.
Timer	Rest timer restores correctly after refresh or 2 minutes in background.
Completion	Every exercise ends with complete-all? yes/no prompt.
Progression	Yes updates progression state according to the attached rule and returns an explanation.
History	After completion, all sets and progression changes are visible in exercise/session history.
Deload	Week 4 automatically generates reduced sets and light planche prescription.
Swim/climb	Week 2/4 Friday and Week 3 Saturday replacements appear automatically.
Security	Authenticated user cannot read another user_id via Supabase client due to RLS.

Area	Pass condition
Offline	At least one full session can be completed with network interrupted after session start and sync later.
Undo	User can undo progression from session summary without deleting the session result.

13.2 Progression unit tests

- Weighted pull-up success at +20 kg -> next state +21 kg.
- Weighted pull-up two consecutive failures -> -2 kg from failed load.
- HSPU 4x5 success -> 4x6; 4x6 -> 4x7; 4x7 -> next level 4x5.
- Planche two successful hard sessions -> advance level and reset hold durations.
- Front split two successful deep/moderate credits -> target gap decreases by 1 cm only.
- Week 4 deload transforms 4 sets to 3 and 3 sets to 2 without changing reps.
- Swim success adds exactly 50 m until 1200 m cap.
- Climbing problem completed in 2 attempts advances grade; 3 attempts holds grade; no completion holds grade.
- Undo restores exact pre-event progression state.

14. Cursor implementation plan and master build prompt

14.1 Recommended build sequence

Phase	Deliverable
1. Foundation	Create Next.js app, Supabase project wiring, Auth, theme, app shell, RLS baseline.
2. Domain model	Migrations for exercise/routine/schedule/session/progression tables; seed exercise library and this plan.
3. Today + Plan	Generate 4-week calendar, Today screen, routine detail view.
4. Session player	Snapshot session, set tracking, exact timer behavior, expandable instructions, offline draft.
5. Progression engine	Server-side rule evaluator + unit tests + progression event audit/undo.
6. History	Session history, exercise history, charts, PR/progression events.
7. Wellbeing	Readiness check-in and pain-aware progression freeze.
8. PWA/polish	Offline queue, installability, wake lock where supported, haptics/sound, accessibility pass.
9. QA	Playwright end-to-end tests against acceptance criteria; seed/demo reset command.

14.2 Cursor master prompt

You are building "Lifetime Athlete", a mobile-first personal training and wellbeing tracker. Treat this PDF as the product and domain source of truth.

Tech constraints:

- Next.js App Router + TypeScript, current stable versions.
- Supabase Postgres + Auth; follow the current official SSR/cookie pattern.
- Tailwind CSS; polished responsive UI; dark mode required.
- Secure all user-owned data with RLS.
- Runtime validation for mutations.
- The training plan, exercise prescriptions and progression logic MUST be data-driven, seeded in Supabase, not hard-coded in React pages.

Core experience:

1. Today shows exactly what is due and starts it immediately.
2. A session guides the user exercise-by-exercise and set-by-set.
3. Exact sets/reps/holds/load/rest are visible at all times.
4. Rest timer auto-starts after set completion and survives background/refresh.
5. Exercise instructions open inline without leaving the session.
6. After every exercise ask whether all prescribed work was completed.
7. Evaluate the attached exercise-specific progression rule server-side.
8. Show the exact next prescription/change and store a progression event.
9. Store all set/session history and expose charts/history.
10. Support the 4-week cycle, deload, swim substitutions and climbing substitution.
11. Preserve active sessions offline and sync later.

Build iteratively in the phases specified in Section 14.1. After each phase:

- run typecheck/lint/tests,
- verify RLS assumptions,
- demonstrate the key user flow,
- do not move workout logic into UI components.

Prioritize excellent one-handed mobile UX and reliability over visual complexity.

14.3 Definition of done for v1

v1 is complete when the user can install the app, open Today, perform every routine in this 4-week plan with exact guidance and timers, have each exercise update its next prescription correctly, review progression/history, complete a full cycle including deload/swim/climbing substitutions, and lose no workout data through refresh or temporary offline use.

Appendix A. Seed plan data model example

The exact SQL/seed implementation can vary. The following illustrates how routine data should look conceptually so Cursor does not encode the plan in JSX.

```
{
  "routine": "Monday - Strength + Power",
  "duration_min": 90,
  "items": [
    {
      "sequence": 1,
      "exercise": "wrist-rocks",
      "block": "warmup",
      "sets": 1,
      "reps": {
        "forward": 10,
        "lateral": 10
      },
      "rest_seconds": 0
    },
    {
      "sequence": 9,
      "exercise": "oahs-practice",
      "block": "skill",
      "duration_seconds": 300,
      "progression_scope": "global_skill",
      "rule": "OAHs_V1"
    },
    {
      "sequence": 10,
      "exercise": "planche-hold",
      "block": "skill",
      "protocol": "hard",
      "duration_seconds": 300,
      "progression_scope": "global_skill",
      "rule": "PLANCHE_V1"
    },
    {
      "sequence": 15,
      "exercise": "weighted-pull-up",
      "block": "strength",
      "sets": 4,
      "reps_per_set": 5,
      "load_from_state": true,
      "rest_seconds": 150,
      "rule": "WEIGHTED_PULLUP_4X5_V1"
    }
  ]
}
```

```

{
  "sequence": 16,
  "exercise": "handstand-push-up",
  "block": "strength",
  "sets": 4,
  "reps_per_set": 5,
  "level_from_state": true,
  "rest_seconds": 150,
  "rule": "HSPU_LEVEL_V1"
}
]
}

```

Each generated session copies routine_items plus resolved current progression values into session_items.prescription_snapshot. Historical data therefore remains reproducible even after progression or plan edits.

Appendix B. Technical references

Use current official documentation during implementation because framework APIs can evolve. The following were current when this specification was created:

Reference	URL
Next.js documentation - App Router	https://nextjs.org/docs
Next.js data mutation / Server Actions concepts	https://nextjs.org/learn/dashboard-app/mutating-data
Supabase Auth with Next.js	https://supabase.com/docs/guides/auth/quickstarts/nextjs
Supabase server-side rendering / @supabase/ssr	https://supabase.com/docs/guides/auth/server-side
Supabase Row Level Security	https://supabase.com/docs/guides/database/postgres/row-level-security
Supabase secure data guidance	https://supabase.com/docs/guides/database/secure-data
Supabase Realtime	https://supabase.com/docs/guides/realtime

Implementation note

Supabase currently recommends its SSR package for cookie-based Next.js server-side auth, but its API can change. Cursor should consult the current official Supabase guide rather than blindly copying an old snippet. RLS must remain enabled regardless of client/server rendering choices.

Safety note

This application is a training tracker, not a medical device. Persistent/worsening pain, radiating symptoms, numbness, weakness or other concerning symptoms should not be solved by automatic exercise progression. Underwater breath-hold training should not be performed alone, and hyperventilation should not be used before breath-holding.